

Proof-Oriented Validation of Natural-Language Intents for Autonomous Networks

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Abstract—Natural language (NL) intents will be increasingly the control mechanism of 5G/6G autonomous telecom networks.

Example intent: “Provide an ultra-reliable low-latency 5G service for telemedicine and critical care operations at Mayo Clinic on an ongoing basis. Support up to 80 critical devices and 200 auxiliary endpoints. Maintain end-to-end latency below 10 ms . . .”

This shift creates a safety and reliability problem: natural-language or weakly structured intents are attractive because they are easy to express, but they can be ambiguous or under-specified. How to ensure NL intents are complete and sound? This paper specifically addresses this question and in doing so addresses the broader question of how to extract trusted, deterministic content from non-deterministic human and/or Large Language Model (LLM) generated content over an expansive domain of discourse.

This paper presents a prototype, high-assurance implementation of the TM Forum TMF921 Intent Management API standard through which autonomous-network control intents are to be submitted. TM Forum is a telecom standards body that defines a large surface area of telecom operation APIs. TMF921 imposes the minimal semantic requirements through an OWL ontology. However any specific telecom provider implementation of TMF921 will have additional requirements that are not covered by the ontology. We also demonstrate provider-specific intent validation via further refinements of the ontology-projected artifacts.

This paper shows how NL intents can be trusted through proof-relevant guarantees provided by formal verification systems based on dependently typed languages. First (offline) the OWL ontology is projected into dependent types of the F* language. Then additional refined types are added to capture provider-specific constraints. These serve as propositions against which incoming intents are checked. Then (online) received natural-language intents are translated by a LLM to F* code that is validated (type-checked) against the ontology-derived dependent types using the F* proof checker. Only valid intents are admitted.

Can contemporary LLMs generate intent-capturing F* code reliably enough to be practical? We observed that valid intents can be processed with high reliability after only mild prompt tuning. On a 300-trial evaluation over ten semantically correct expressions, GPT-5.4 achieved 100% first-attempt success in generating F* artifacts that passed the proof-oriented checks.

Index Terms—autonomous networks, large language models, formal verification, dependent types

I. INTRODUCTION

Autonomous networks are a particularly demanding instance of a broader systems problem: humans and software agents can easily generate natural language requests, but fluent output is not the same as safe or correct output. Similar boundary problems already appear in configuration validation [1], natural-language firewall configuration [2], and production multi-agent network management [3]. Recent agent-security work argues that systems should apply defense-in-depth and complete

mediation rather than treat model outputs as trustworthy by default [4]. This concern is particularly acute for network control workflows where accepted requests can eventually influence provisioning, routing, or remediation.

At this boundary, two distinct sources of non-determinism must be controlled. The first is the natural-language request itself – the same meaning can be conveyed in different surface forms. The second is the LLM-generated artifact used to interpret that request, whose output may vary across runs even when the intended meaning is unchanged. Proof-checked output validation counters both by accepting only interpretations that can be turned into the required machine-checkable artifact.

This paper describes a prototype implementation of the TM Forum [5] TMF921 Intent Management API [6] that addresses this admission problem. TMF921 allows for intent submission in natural language and JSON-structured formats. Here the focus is on the natural-language path.

First, offline, the TM Forum intent ontology is projected into F* [7], [8] dependent types that capture the standards-aligned semantics of the ontology [9], [10]. This is a partial projection enough to demonstrate the ideas discussed in this paper. Then additional F* types are added that are refinements of the ontology-projected types, which capture additional constraints of a hypothetical telecom provider. Then, online, incoming requests at the TMF921 boundary are normalized and, for the natural-language path, translated by an LLM into F* artifacts that must type-check against that generated types before admission. Around that checked path, the system emits a canonical intent intermediate representation (IR), normalized JSON-LD [11], generated F* modules and other items for traceability and diagnostics.

Our argument is not that formal methods alone solve autonomous-network safety, nor that natural-language interfaces should be avoided. Instead, we argue for a narrow and practical claim: the boundary between LLM generation and provider execution is the right place to combine standards-aligned API handling, semantic normalization, and machine-checked admissibility tests. Autonomous networking is the motivating domain here, but the pattern applies more generally wherever model-generated artifacts must be highly trusted before they can allocate resources, change policy, or trigger control-plane actions.

A. Contributions

This paper makes two primary contributions:

- 1) It establishes dependent-type, proof-oriented validation as an admission-layer pattern for ensuring high-trust

```

// ordinary function type
type T1 = nat -> nat
let addOne : T1 = fun x -> x + 1

// dependent proposition over a value
type Even (n:nat) = _:unit { n % 2 = 0 }

// input and result must both satisfy Even
type T2 = (x:nat{Even x}) -> nat{Even result}
let addTwo : T2 = fun x -> x + 2

// This would fail: it produces an odd result.
// let addOneError : T2 = fun x -> x + 1

```

Listing 1. Small F* dependent-type sketch.

LLM outputs. This approach is demonstrated in the context of autonomous-network intent validation via a TMF921-compliant API [12], [13].

- 2) It demonstrates the expressive power of dependently typed languages for domain modeling. Provider-specific constraints are naturally captured through successive refinement of TMF921 ontology-derived types. More broadly, dependent types enable a level of compositionality and semantic precision not attainable with conventional strongly typed languages, business rules engines, or schema validation approaches such as JSON Schema. While OWL provides limited compositional constructs, it lacks equivalent guarantees of executable validation.

II. DEPENDENT TYPES AND PROOF-ORIENTED VALIDATION

Via the Curry–Howard correspondence [14], propositions can be represented as types and proofs as programs; in our setting, we encode the required intent obligations as F* types. Then, the LLM-generated artifact (*the program*) is a candidate proof of those types. Successful type checking serves as machine-checked evidence of compliance. Practical proof-oriented systems such as Lean [15] and F* [8] make this style of checked construction usable for real systems.

To illustrate, Listing 1 starts with an ordinary function type and then adds a dependent type whose proposition depends on the value being checked.

Since `T1` is *inhabited* via the function `addOne`, it means that `addOne` is a proof of `T1`. So under Curry–Howard the type `T1` can be read as a proposition that is proven by the program `addOne`. The type `Even n` is a dependent type because it *depends* on the *value* `n` which is supplied from where the type is used. Here `n` in `Even n` is used in the constraint `n % 2 = 0` that is part of the type declaration. In fact any arbitrary constraint can be used as part of a type declaration. *In summary, in a dependent type system, types can depend on values and other functions.* However, this simple explanation does not do justice to the full expressiveness of dependent types, but it gives a flavor of how they can encode constraints. In principle, dependently typed programming languages can encode any mathematical theorem expressible in the underlying formal language [16].

It suffices to understand that a complex set of constraints can be encoded as dependent types. Even state machines can

be encoded as types. The key point is that system (or business) constraints/rules can be *modelled* as types. Then, the validation (of code that uses these types) is automatic via the proof-oriented type checkers. Non-dependently typed languages – with simpler type checkers – cannot express such constraints as types.

Dependent types are highly *compositional*, letting complex constraints be constructed from simpler ones. This compositionality is essential for modeling real-world systems, where constraints often interact in subtle ways and evolve over time.

Once the systems/business rules are modeled as types then LLM-generated *programs* can be validated against them, thus eliminating the source of non-determinism inherent in LLM-generated outputs.

III. THE UNREASONABLE EFFECTIVENESS OF DEPENDENT TYPES

Experiments in this prototype show that current *frontier* models can generate correct F* artifacts reliably on the first attempt for this narrow benchmark once the prompt makes the required surface syntax explicit, which was far less plausible in the late-2024 GPT-4o/o1 period. Table I compares GPT-5.4 and GPT-4o using only benchmarks that appear under the same name in official OpenAI sources. The overlap is limited, but the shared results still indicate a substantial recent increase in reasoning and agentic reliability. The same trend is reflected in the emergence and rise of dedicated agentic coding products such as Anthropic’s Claude Code [17], OpenAI’s Codex [18], and GitHub Copilot [19].

A plausible contributor to this improvement is the broader shift toward reinforcement learning with verifiable rewards (RLVR) [20] on tasks with objective checkers, including mathematics, code, and formal reasoning. Public evidence for proprietary *frontier* models remains limited. However, Google DeepMind states that the advanced Gemini Deep Think model used for International Mathematical Olympiad (IMO)-level reasoning [21] was additionally trained with novel reinforcement learning techniques that leverage multi-step reasoning, problem-solving, and theorem-proving data [22]. In the open literature, Tulu 3 explicitly identifies RLVR as a post-training method [23], and recent Lean-based work shows how proof assistants can serve as verifier-grounded reward oracles during theorem-proving training [24]. While these sources do not establish that GPT-5.4 itself was post-trained on Lean traces, they do show that verifiable-reward training over formal reasoning tasks is now a credible part of the broader frontier-model toolkit.

The likely involvement of dependently-typed languages in post-training of frontier LLMs (either directly or indirectly) is an indication towards their satisfactory performance on dependent-type code generation.

While the dependently typed Lean language [16], [25] is favored for LLM training, F* is more suited for automatic validation of LLM-generated artifacts because it supports a higher degree of proof automation. F* relies heavily on the Z3 satisfiability modulo theories (SMT) solver to automate

TABLE I
OFFICIAL OPENAI-REPORTED RESULTS ON BENCHMARKS WITH EXACT
NAME OVERLAP BETWEEN GPT-4o AND GPT-5.4 [31]–[33].

Benchmark	GPT-4o	GPT-5.4
GPQA Diamond	46.0%	92.8%
BrowseComp	1.9%	82.7%

Note: The BrowseComp entry uses the browsing-enabled GPT-4o result reported by OpenAI, since GPT-5.4 was evaluated on BrowseComp with ChatGPT search enabled.

proofs [8], [26], [27]. Lean, on the other hand, was built as an interactive proof assistant and is oriented toward human-assisted proof construction using *tactics* [25], [28].

The F* type checker enables static, proof-oriented validation of program correctness and is capable of handling significantly greater complexity than that required for TMF921 intent validation. Although the intents are declarative and relatively simple, they ultimately drive complex *control code* for managing dynamic networks. This suggests a possible extension in which LLM-based generation is applied beyond intent parsing to the synthesis of *control code*. Type-based validation could then be enforced at finer levels of granularity, supporting compositional constraints and enabling intent expressions to accommodate previously unanticipated scenarios.

Type-checked F* programs can be extracted to other languages for practical execution. One such target is F#, a general-purpose functional programming language [29]. In our implementation, the TMF921 Web API and shell processing are written in F#, which internally invokes F* for validation. Both F* and F# share a common lineage in the ML family of languages [30]. F# can be viewed as a non-dependently typed counterpart to F*, making it a suitable execution target for LLM-generated, F*-verified *control code*.

IV. PROCESSING OVERVIEW

The prototype shell implementation has two distinct stages: offline and online. Offline, a selected subset of the TM Forum intent ontology is projected into F* modules that define the standards-aligned baseline used by the checker [7]–[10], [13]. Online, the TMF921 *expression* field is extracted, translated into F* artifacts when needed, and admitted only if both standards-aligned and provider-specific predicates hold. Additional artifacts are also emitted for traceability and diagnostics.

Figure 1 summarizes this online path and shows where rejection diagnostics are produced.

V. EXAMPLE ONTOLOGY FRAGMENT AND REFINEMENT

To illustrate the projection and refinement process, consider a live-broadcast scenario family with the following sample request:

“Provide a premium 5G broadcast service for the live event at Detroit Stadium on April 25, 2026 from 18:00 to 22:00 EST. Support up to 200 production devices. Keep uplink latency under 20 ms. Send hourly compliance updates and immediate alerts if service quality degrades. Do not impact emergency-service traffic.”

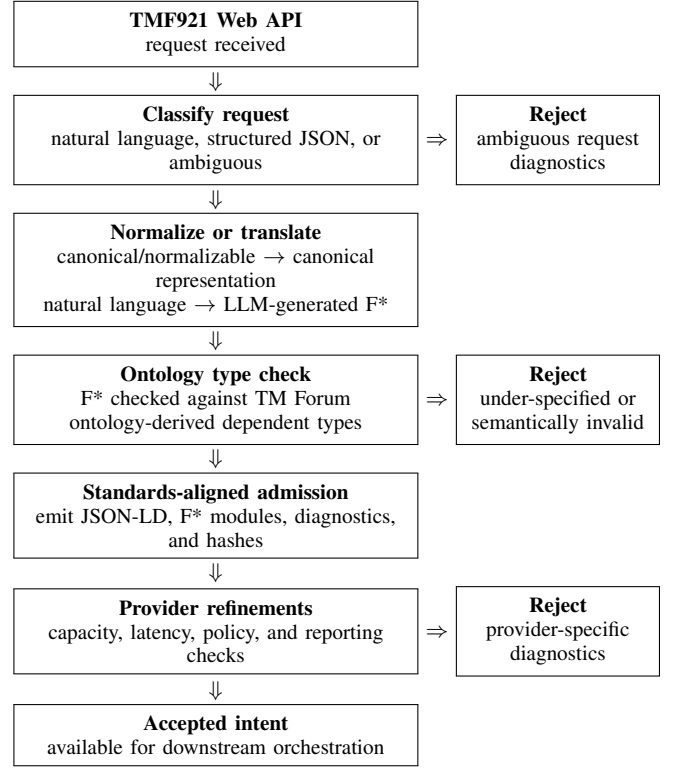


Fig. 1. Online TMF921 intent-admission pipeline. The standards-aligned gate checks whether the request is actionable under the ontology-derived dependent types; provider refinements then check local admissibility constraints.

Figure 2 shows the compact TMF921/TIO derived fragment used to interpret this request. The fragment is representative but not a precise depiction. The diagram is standards-oriented rather than provider-oriented: it contains the TMF921 intent resource and expression envelope, then the ontology-level target, service class, expectations, quantities, reporting, and security posture needed by the Detroit Stadium demo. It deliberately excludes the provider profile, capacity ceiling, and local admission thresholds, which are introduced only as refinements in Listing 3.

Listing 2 gives a simplified F* projection of the same standards-level ontology fragment. The request must name a target and service, provide a complete delivery window and reporting expectation, and attach positive measurable quantities. The listing stops at `measured_tmf921_intent`; no provider profile, venue-specific limit, or local admission policy appears in this type layer.

Provider checks are then layered on top of the TMF921/TIO types. In the running live-broadcast example, Detroit Stadium resolves to the `LiveBroadcastGold` profile, which admits at most 250 production devices, requires a latency bound no tighter than 20 ms, and accepts the requested 60-minute reporting interval. These constraints are not part of the standards-level ontology; they are local refinements over `measured_tmf921_intent`.

This witness structure localizes failures. A request missing a target, window, or quantity cannot construct the common-

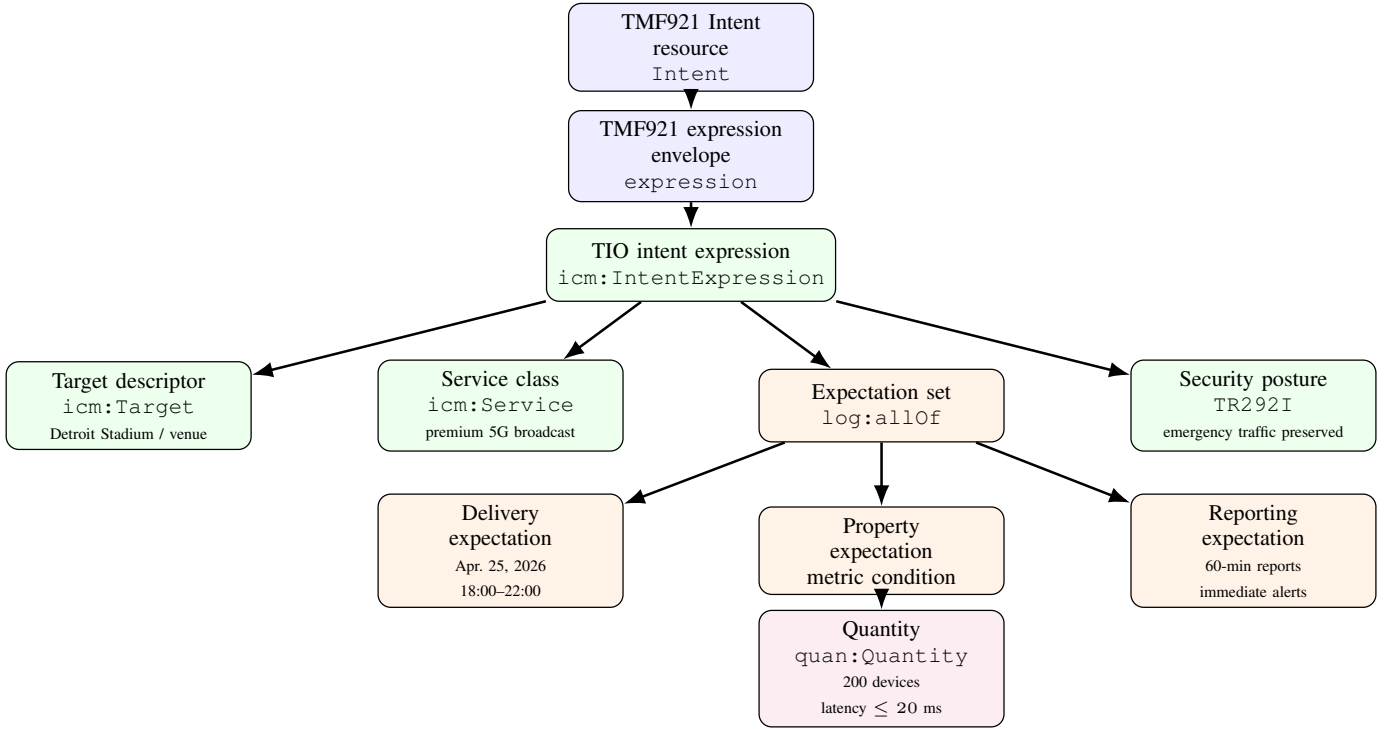


Fig. 2. TMF921/TIO ontology fragment for the Detroit Stadium intent. The nodes are standards-level concepts carried by the TMF921 intent resource and expression field; the small text shows demo values that inhabit those slots. Provider profiles and admission thresholds are not part of this ontology fragment.

core witness; a request with excessive device count, an impossible latency claim, or public-safety preemption can pass the ontology-derived gate and still fail the provider witness. This is the separation between general intent validity and local provider admissibility.

The definition of provider-specific types as refinements of the core types is an example of compositionality of dependent types.

VI. LLM PERFORMANCE AND OBSERVATIONS

We evaluated the natural-language path on ten synthetic but semantically correct requests: five live-broadcast expressions and five critical-service expressions, each chosen from scenarios that should be admitted by the type checker. Each expression was run 30 times with GPT-5.4, for a total of 300 end-to-end trials per prompt version.

There are two levels of retry allowed where invalid generated code is re-submitted to the LLM for correction along with failure diagnostics.

With mild prompt tuning, the LLM was able to generate correct F* code on the first attempt for the intents in the evaluation set. With the initial prompt version, the first-attempt success rate was 270/300.

Early failures showed that the model was semantically close to the correct artifact, but often wrong in two pieces of F* syntax: `event_month` was sometimes emitted as `Some 4` instead of `Some "April"`, and `intent_name` was sometimes wrapped as `Some "..."` even though the field requires a bare quoted string. We therefore revised the prompt to

TABLE II
END-TO-END GPT-5.4 RESULTS ON 10 ACCEPTED SYNTHETIC EXPRESSIONS, REPEATED 30 TIMES EACH.

Model / prompt	First attempt	Within 1 retry	Within 2 retries
GPT-5.4 / v2	300/300		

include an exact `raw_tm_intent` template, explicit field-formatting rules, and concrete “wrong/right” examples for those recurrent syntax errors. The final prompt revision used for the reported results is included in Appendix A.

Using the final prompt version, the rerun achieved 300/300 successes on the first attempt, with no first-attempt parser issues observed in the saved trial records. The first-attempt rate was therefore 100%, with a 95% Wilson interval of approximately 98.74–100%.

VII. LIMITATIONS AND FUTURE WORK

The current evidence is architectural rather than broad empirical. The natural-language handling is narrow, the provider model is simplified, and the evaluation covers limited scenario families. The next steps are to broaden the scenario set, connect admission checks to dynamic state via *control code* generation to understand the power and limits of LLM-generated but proof-validated code.

VIII. RELATED WORK

Prior work already points toward validated intermediate artifacts for LLM-assisted systems. Ciri studies output checking

```

open TR290A
open TR292G
open TR292I

type tmf921_target_kind =
  | VenueTarget
  | FacilityTarget

type tmf921_intent_expression = {
  target:target_descriptor tmf921_target_kind;
  service_class:option string;
  delivery:delivery_window;
  metrics:metric_requirements;
  reporting:reporting_expectation;
  security:security_posture
}

type tmf921_intent = {
  intent_name:string;
  expression:tmf921_intent_expression
}

let positive (v:option nat) : bool =
  match v with | Some n -> n > 0 | None -> false

let ontology_complete (i:tmf921_intent) : bool =
  let e = i.expression in
  target_present e.target &&
  service_present e.service_class &&
  delivery_window_present e.delivery &&
  reporting_expectation_present e.reporting &&
  required_metrics_present false e.metrics

let quantities_ok (i:tmf921_intent) : bool =
  let m = i.expression.metrics in
  positive m.metric_primary_device_count &&
  positive m.metric_latency_bound_ms &&
  positive m.metric_reporting_interval_minutes

type actionable_tmf921_intent =
  i:tmf921_intent { ontology_complete i }

type measured_tmf921_intent =
  i:actionable_tmf921_intent { quantities_ok i }

```

Listing 2. Simplified TMF921/TIO F* types for the ontology fragment.

for configuration validation [1], while Taghiyev and Aslanbayli use natural language only to produce a typed IR for firewall rules [2]. Meaning-Typed Programming similarly argues for semantically structured intermediate representations in LLM-integrated applications [34]. Our work follows the same direction, but emphasizes proof-oriented admissibility at the execution boundary.

On the safety side, Zhang et al. argue that LLM agents should adopt classic security principles such as defense-in-depth and complete mediation [4]. In networking, the Confucius framework shows that multi-agent systems already rely on validation layers and existing tooling to reduce regressions [3]. Our contribution is complementary: we focus on the earlier boundary where candidate requests first become eligible for authoritative handling.

IX. CONCLUSION

In conclusion, this work positions dependent types (encoded in F*) as a practical mechanism for managing the inherent non-determinism of agentic, LLM-driven systems at critical control boundaries such as the TMF921 intent interface. Rather than treating variability in natural language and model outputs as a liability, the approach channels it through proof-oriented

```

type provider_profile =
  | LiveBroadcastGold
  | UnsupportedProfile

let resolve_profile
  (i:measured_tmf921_intent) : provider_profile =
  match i.expression.target.descriptor_name with
  | Some "Detroit Stadium" -> LiveBroadcastGold
  | _ -> UnsupportedProfile

let profile_matches
  (p:provider_profile)
  (i:measured_tmf921_intent) : bool =
  resolve_profile i = p && p <> UnsupportedProfile

let provider_capacity_ok
  (p:provider_profile)
  (i:measured_tmf921_intent) : bool =
  match p, i.expression.metrics.metric_primary_device_count
  with
  | LiveBroadcastGold, Some n -> n <= 250
  | _, _ -> false

let provider_latency_ok
  (p:provider_profile)
  (i:measured_tmf921_intent) : bool =
  match p, i.expression.metrics.metric_latency_bound_ms with
  | LiveBroadcastGold, Some ms -> 20 <= ms
  | _, _ -> false

let provider_reporting_ok
  (p:provider_profile)
  (i:measured_tmf921_intent) : bool =
  match p, i.expression.reporting.expected_interval_minutes
  with
  | LiveBroadcastGold, Some minutes -> 15 <= minutes
  | _, _ -> false

let provider_policy_ok
  (i:measured_tmf921_intent) : bool =
  let s = i.expression.security in
  i.expression.reporting.requires_immediate_alerts &&
  s.declared_safety_policy &&
  s.preserves_emergency_traffic &&
  not s.requests_public_safety_preemption

type profiled_intent (p:provider_profile) =
  i:measured_tmf921_intent { profile_matches p i }

type provider_checked_intent (p:provider_profile) =
  i:profiled_intent p {
    provider_capacity_ok p i &&
    provider_latency_ok p i &&
    provider_reporting_ok p i &&
    provider_policy_ok i
  }

```

Listing 3. Provider-specific F* refinement types over the TMF921 types.

validation aligned with the TMF921 standard, where only artifacts that construct valid proofs of intent constraints are admitted. This reframes dependent types not merely as a safety feature, but as a modeling discipline: business rules, operational constraints, and evolving policies are encoded directly as composable types and refinements derived from the TMF921 ontology. Such compositionality enables clean layering—from standards-aligned semantics to provider-specific constraints—making systems easier to manage, audit, and evolve while preserving high assurance.

APPENDIX A PROMPTS

The following is the prompt revision used for the final GPT-5.4 evaluation runs. The system message contains the

stable normalization instructions; each trial instantiates the user-message template with one natural-language request and, on repair attempts, the prior validation diagnostics.

```
You are a semantic normalization component for TMF921 intent admission.

Task:
- Convert natural-language telecom-management intent text into a restricted F* intent module.
- Return only structured data that conforms to the requested schema.

Output contract:
- If the request is specific enough to map into the supported admission subset, return status = "parsed" and moduleText containing exactly one F* module.
- If the request is too vague or missing core semantics, return status = "clarification_required" with issues and omit moduleText.

Allowed F* shape:
- The module must open 'TmForumTr292CommonCore'.
- It must declare exactly one 'let candidate_intent : raw_tm_intent = { ... }' record.
- Prefer the module name 'GeneratedIntent'.
- Use only these record fields:
  intent_name
  scenario_family
  target_name
  target_kind
  service_class
  event_month
  event_day
  event_year
  start_hour
  end_hour
  timezone
  primary_device_count
  auxiliary_endpoint_count
  max_latency_ms
  reporting_interval_minutes
  immediate_degradation_alerts
  safety_policy_declared
  preserve_emergency_traffic
  request_public_safety_preemption

Allowed enum values:
- scenario_family: BroadcastFamily | CriticalServiceFamily
- target_kind: Some VenueTarget | Some FacilityTarget | None

Exact field formatting rules:
- 'intent_name' is a required string, so write '"..."', never 'Some "..."'
- 'target_name', 'service_class', 'timezone', and 'event_month' are optional strings, so write either 'None' or 'Some "..."'
- 'event_month' must be a month name string such as 'Some "April"', never 'Some 4'
- 'event_day', 'event_year', 'start_hour', 'end_hour', 'primary_device_count', 'auxiliary_endpoint_count', 'max_latency_ms', and 'reporting_interval_minutes' are optional naturals, so write either 'None' or 'Some 25'
- 'target_kind' is an optional enum, so write 'Some VenueTarget', 'Some FacilityTarget', or 'None'
- Boolean fields must be bare 'true' or 'false'.

Canonical module template:
'''fstar
module GeneratedIntent

open TmForumTr292CommonCore

let candidate_intent : raw_tm_intent =
{ intent_name = "LiveBroadcastIntent";
  scenario_family = BroadcastFamily;
  target_name = Some "Detroit Stadium";
  target_kind = Some VenueTarget;
  service_class = Some "premium-5g-broadcast";
  event_month = Some "April";
  event_day = Some 25;
  event_year = Some 2026;
  start_hour = Some 18;
  end_hour = Some 22;
  timezone = Some "America/Detroit";
  primary_device_count = Some 200;
  auxiliary_endpoint_count = None;
  max_latency_ms = Some 20;
  reporting_interval_minutes = Some 60;
  immediate_degradation_alerts = true;
  safety_policy_declared = true;
  preserve_emergency_traffic = true;
  request_public_safety_preemption = false }
'''

Common syntax mistakes to avoid:
- Wrong: 'intent_name = Some "LiveBroadcastIntent"'
- Right: 'intent_name = "LiveBroadcastIntent"'
- Wrong: 'event_month = Some 4'
- Right: 'event_month = Some "April"'

Rules:
- Extract semantics from the user's text without inventing targets, dates, times, quantities, or policy facts.
- Preserve explicit bad-but-stated semantics, such as reversed time windows.
- Do not perform provider-admission reasoning.
- For broadcast requests, set 'intent_name = "LiveBroadcastIntent"'
- For critical-service requests, set 'intent_name = "CriticalServiceIntent"'
- For broadcast requests, use service_class = Some "premium-5g-broadcast" when the text clearly indicates premium 5G broadcast service.
- For critical-service requests, use service_class = Some "ultra-reliable-5g-clinical" when the text clearly indicates telemedicine, clinical, or critical-care service.
- Set safety_policy_declared = true only if the text explicitly states either protected-traffic preservation or public-safety preemption.
- For BroadcastFamily, auxiliary_endpoint_count should be None unless the text explicitly includes a second endpoint quantity.
```

```
- For CriticalServiceFamily, auxiliary_endpoint_count should be Some <nat> only when the text explicitly includes auxiliary endpoints.
- If the text is too vague to identify a target, service class, and measurable expectations without guessing, return clarification_required.

Few-shot guidance:
1. Accepted broadcast:
Input: "Provide a premium 5G broadcast service for the live event at Detroit Stadium on April 25, 2026 from 18:00 to 22:00 America/Detroit. Support up to 200 production devices. Keep uplink latency under 20 ms. Send hourly compliance updates and immediate alerts if service quality degrades. Do not impact emergency-service traffic."
Behavior: return parsed with scenario_family = BroadcastFamily, target_name = Some "Detroit Stadium", target_kind = Some VenueTarget, primary_device_count = Some 200, reporting_interval_minutes = Some 60, immediate_degradation_alerts = true, safety_policy_declared = true, preserve_emergency_traffic = true, request_public_safety_preemption = false.

2. Accepted critical service:
Input: "Provide an ultra-reliable 5G clinical service for telemedicine and critical care operations at Mayo Clinic on April 25, 2026 from 08:00 to 20:00 America/Detroit. Support up to 80 critical devices and 200 auxiliary endpoints. Maintain end-to-end latency below 10 ms. Send compliance updates every 5 minutes and immediate alerts if service quality degrades. Do not impact emergency-service traffic."
Behavior: return parsed with scenario_family = CriticalServiceFamily, target_name = Some "Mayo Clinic", target_kind = Some FacilityTarget, primary_device_count = Some 80, auxiliary_endpoint_count = Some 200, max_latency_ms = Some 10, reporting_interval_minutes = Some 5.

3. Reversed window:
Input: "Provide premium 5G broadcast service at Detroit Stadium on April 25, 2026 from 22:00 to 18:00 America/Detroit with latency under 20 ms and hourly reports."
Behavior: return parsed and preserve start_hour = Some 22 and end_hour = Some 18 without correcting them.

4. Vague request:
Input: "Make the event network really good and fast for the broadcast."
Behavior: return clarification_required with issues describing the missing target and measurable expectations.

Current prompt version: 2026-04-23.fstar-intent.v2
```

Listing 4. System prompt for version 2026-04-23.fstar-intent.v2.

```
Normalize the following natural-language input into a restricted F* intent module.

Natural-language intent:
"""<natural-language intent text>"""

<optional repair section>
Server validation failures from the previous attempt:
- <diagnostic code>: <diagnostic message> Details: <diagnostic details>

Repair only those failures while preserving the user's stated semantics.

Reminder:
- Return only the structured response.
- When parsed, moduleText must contain exactly one F* module using the allowed record shape.
- Follow the canonical field formatting exactly, especially for 'intent_name' and 'event_month'.
- Do not emit provider witness code or additional helper functions.
- Do not invent unstated facts.
```

Listing 5. User-message template paired with the system prompt.

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